20260223_Progress_Report_XX

Xiaoni Xu

2026-02-23

Overview of Recent Work

In the most recent phase of the project, I refactored the analysis pipeline to directly utilize the `cmcrhazard` R package. This transition from local function definitions to a formal package structure ensures greater reliability and easier maintenance.

Key updates include: - **Package Integration**: Successfully migrated the script to use `cmest_pathcomprisk` from the `cmcrhazard` package. - **Improved Power**: Increased the bootstrap samples to 1,000 and the sample size to 5,000 to obtain more stable estimates. - **Variable Alignment**: Renamed arguments to align with package conventions (e.g., using `D` for dropout/competing risk models).

Model Definitions

The analysis utilized the following model structures for the mediators, competing risks, and outcome:

```r
# Mediator models (mreg)
mreg1 <- lm(M_1 ~ E + C, data = df_main)
mreg2 <- lm(M_2 ~ E + M_1 + C, data = df_main)
mreg <- list(mreg1, mreg2)

# Dropout/Survival models (D) - Indirect Effect through Dropout death
D1_mod <- glm(D_1 ~ E + M_1 + C, data = df_main, family = binomial())
D2_mod <- glm(D_2 ~ E + M_2 + C, data = df_main, family = binomial())
D <- list(D1_mod, D2_mod)

# Outcome model (yreg - Aalen additive hazards)
yreg <- aalen(Surv(tstart, tstop, endpt) ~ const(E) + const(M) + const(C),
    data = df_process, resample.iid = 1, n.sim = 0
)
```

Mediation Analysis Results

The analysis was performed on a simulated dataset with a sample size of $N = 5,000$ and $1,000$ bootstrap iterations.

Effect	Estimate (95% CI)
Direct Effect (DE)	67.59 (-39.2, 164.7)
Indirect Effect through Mediator (IEM)	33.07 (-52.21, 118.88)
Indirect Effect through Dropout death (IED)	-0.32 (-0.81, 0.17)

Effect	Estimate (95% CI)
Total Effect (TE)	99.75 (29.62, 168.63)

Interpretation

Based on the analysis results: - The **Total Effect (TE)** is positive and statistically significant, with an estimate of 99.75. - The path-specific effects indicate a positive direction, suggesting that the mediators partially explain the effect of the treatment on the outcome. - We also observe a small negative effect through the competing risks (**Indirect Effect through Dropout death**).

Next Steps

- Run the analysis with a larger sample size ($N = 7,500$ or $N = 10,000$) to further tighten the confidence intervals for the path-specific effects.
- Continue testing the package stability with more complex simulation scenarios and real-world mimicking datasets.